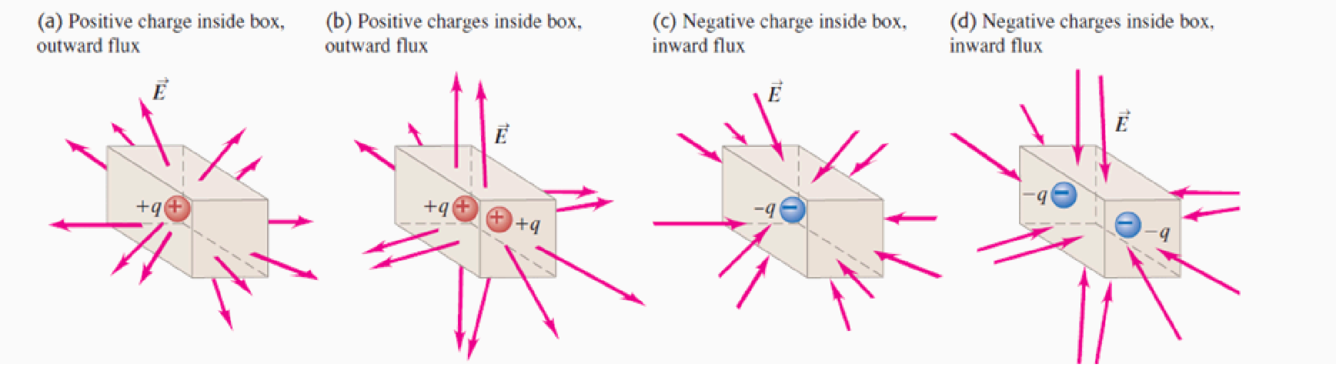
Chapter 23 Gauss' Law

Electric Flux and Enclosed Charge

Electric flux: Number of electric field lines passing through the curved surface.

The electric field vectors point out of the surface, we say there is an outward electric flux. If point into the surface, the electric flux is inward.

22.2 The electric field on the surface of boxes containing (a) a single positive point charge, (b) two positive point charges, (c) a single negative point charge, or (d) two negative point charges.



Whether there is a net outward or inward electric flux through a closed surface depends on the sign of the enclosed charge. Outside the surface do not give a net electric flux since it is through the surface.

The net electric flux proportional to the net amount of charge enclosed but is independent of the size and shape of the closed surface.

Calculating Electric Flux

Flat Surface, Uniform Field

The symbol: Φ_E.

We define the electric flux through this area to be the scalar product of the electric field and the area vector.

$$\Phi_E = \vec{E} \cdot \vec{A}$$

For the area vector

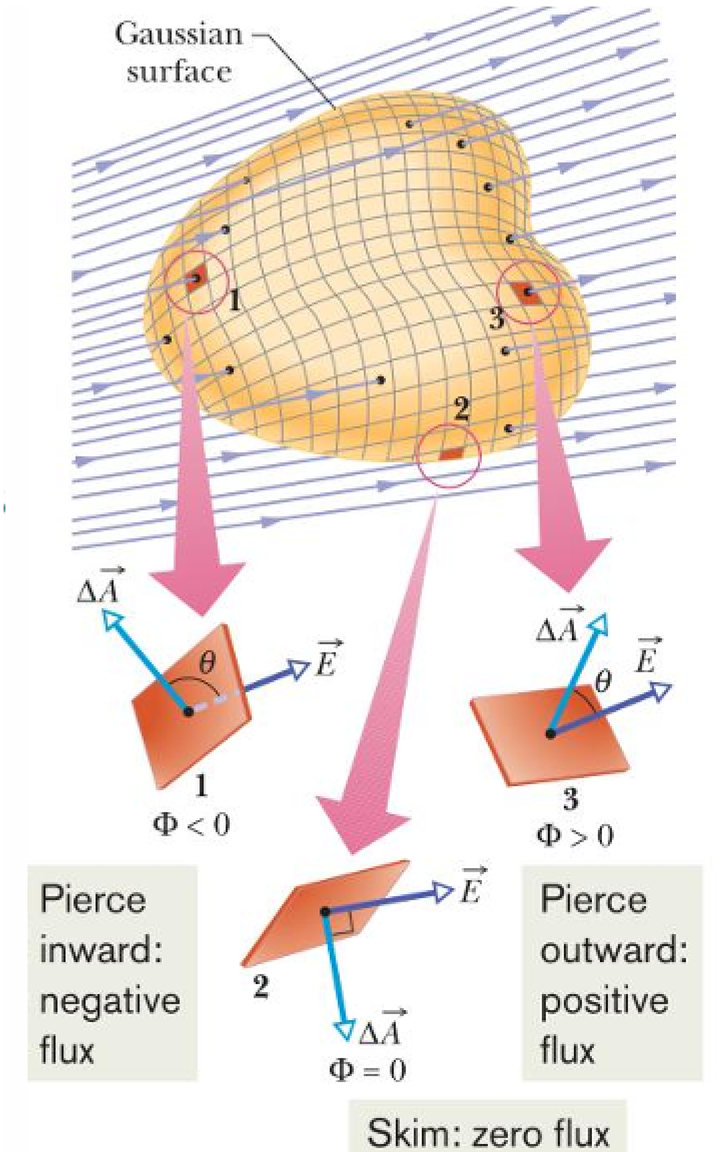
$$\vec{A} = A\hat{n}$$

where $\hat{n}$ is the unit normal vector to the area. With a closed surface, we always choose the direction of $\hat{n}$ to be outward. Thus we call outward electric flux corresponds to a positive value of Φ_E.

Flux of a Nonuniform Electric Field

We divide A into many small element dA, each of which has a unit vector $\hat{n}$ perpendicular to it and $d\vec{A} = \hat{n} dA$. And the total net flux:

$$\Phi = \oint_S \vec{E} \cdot \mathrm{d}\vec{A}$$



Gauss's Law

It is an alternative to Coulomb's Law. Provide a different way to express the relationship between electric charge and electric field.

Gauss's Law:

$$\Phi_E = \oint_S \vec{E} \cdot d\vec{A} = \frac{q_{enc}}{\epsilon_0}$$

where q_{enc} is the algebraic sum of net charge enclosed by the closed surface(Gaussian surface), E is total electric field including both inside and outside. The surface S is called the Gaussian surface.

Problem Solving Strategy using Gauss's Law

- Analyze the symmetry of charge distribution and the electric field.
- Establish a suitable coordinate system
- Select a suitable Gaussian Surface
- Apply Gauss's Law to calculate the electric field strength.